AP CALCULUS AB

Unit 1 • Lesson 1.14 practice • Answer key

Lesson 1.14 • Connecting Infinite Limits and Vertical Asymptotes

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 0^-} \frac{1}{x}$ is

(A) $+\infty$

(B) $-\infty$

(C) 0

(D) 1

Solution

Answer: (B). For small negative x, $\frac{1}{x}$ is a large negative number.

Question 2

NO CALCULATOR

$\lim_{x \to 2} \frac{3}{(x-2)^2}$ is

(A) $+\infty$

(B) $-\infty$

(C) 3

(D) It does not exist and is not infinite.

Solution

Answer: (A). The denominator is positive on both sides of 2 and approaches 0.

Question 3

NO CALCULATOR

Which line is a vertical asymptote of $f(x) = \ln(x-1)$?

(A) $x = 0$

(B) $x = e$

(C) $x = 1$

(D) $y = 1$

Solution

Answer: (C). $\lim_{x \to 1^+} \ln(x-1) = -\infty$.

Question 4

NO CALCULATOR

$\lim_{x \to \frac{\pi}{2}^+} \tan x$ is

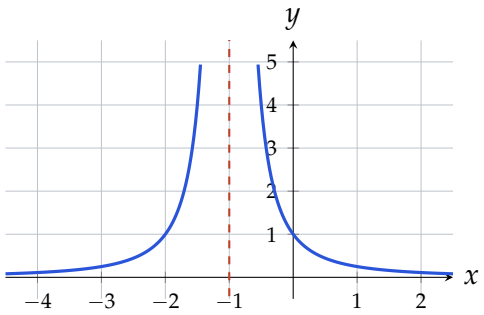
(A) $+\infty$
(B) 0
(C) 1
(D) $-\infty$

Solution
Answer: (D). Just to the right of $\frac{\pi}{2}$, $\sin x > 0$ and $\cos x < 0$ is near 0, so $\tan x \to -\infty$.

Question 5 (graph)

NO CALCULATOR

The graph of $f(x) = \dfrac{1}{(x+1)^2}$ is shown. Which describes $\lim_{x \to -1} f(x)$?



(A) $+\infty$
(B) $-\infty$
(C) 0
(D) 1

Solution
Answer: (A). The graph rises without bound on both sides of the asymptote $x = -1$.

Question 6 (Free response)

NO CALCULATOR

Let $f(x) = \dfrac{x+2}{x^2 - 4x}$.

(a) Find all vertical asymptotes of the graph of f.
(b) For each asymptote, find both one-sided limits. Justify the signs.

Solution
(a) $f(x) = \dfrac{x+2}{x(x-4)}$. No factor cancels: $x = 0$ and $x = 4$.
(b) Near $x = 0$ the numerator is near 2 and $x - 4$ near -4. For $x < 0$, $x(x-4) > 0$: $\lim_{x \to 0^-} f(x) = +\infty$. For $x > 0$, $x(x-4) < 0$: $\lim_{x \to 0^+} f(x) = -\infty$.
Near $x = 4$ the numerator is near 6 and x near 4. For $x < 4$, $x(x-4) < 0$: $\lim_{x \to 4^-} f(x) = -\infty$. For $x > 4$: $\lim_{x \to 4^+} f(x) = +\infty$.